

Sagemaker pipelines MLOps process

- 1) data validation - great expectations
- 2) data cleaning – pandas
- 3) data validation [data drift, model drift, concept drift, data distribution drift]- tfdv
- 4) data transformation – tft
- 5) model training
- 6) model evaluation
- 7) model deployment

✅ **SageMaker Pipelines** can orchestrate all these MLOps steps! SageMaker Pipelines are highly flexible and can chain together any custom logic using Python containers (or built-in processing/ training steps), so you can integrate **Great Expectations, Pandas, TFDV, TFT, model eval, and deploy** seamlessly.

Here's **how you can structure it** inside a **SageMaker Pipeline**:

1 Data Validation (Great Expectations)

- **Create a ProcessingStep** with a **custom Docker container** or **Python script** that:
 - Installs Great Expectations
 - Loads your data
 - Runs validation suites
- Fail the pipeline if validation fails, or output validation reports to S3.

2 Data Cleaning (Pandas)

- Another **ProcessingStep**:
 - A **processing script** with Pandas to clean the data.
 - Save cleaned output to S3 as CSV/Parquet.

3 Drift Detection (TFDV)

- Another **ProcessingStep**:

- Use TFDV to generate statistics & detect **data drift**, **skew**, and **anomalies**.
- Write summary reports back to S3.
- Condition steps to alert or branch if drift is detected.

⚙️ 4 Data Transformation (TFT)

- **TrainingStep** or **ProcessingStep**:
 - Apply TFT to transform raw features into the training-serving representation.
 - Save transform_graph for reuse at inference.

📦 5 Model Training & Evaluation

- **TrainingStep**:
 - Train your model with TensorFlow/Keras.
- **ProcessingStep**:
 - Run model evaluation scripts (e.g. computing accuracy, perplexity, etc.).
 - Integrate **TFMA** if you want slicing and more granular metrics.

🚀 6 Deployment

- **Register the model** in **SageMaker Model Registry**.
- **Create a Model and deploy** to a **SageMaker Endpoint** with:
 - Autoscaling
 - Model monitoring enabled

🌱 Diagram of the SageMaker Pipeline Steps

[Great Expectations Validation] → [Pandas Cleaning] → [TFDV Drift Check] → [TFT Transform] → [Train] → [Evaluate (TFMA/Custom)] → [Register Model] → [Deploy to Endpoint]

✔ **Technologies you'll use:**

Step	Tool	SageMaker Step Type
Data validation	Great Expectations	ProcessingStep
Data cleaning	Pandas	ProcessingStep
Drift detection	TFDV	ProcessingStep
Transformation	TFT	ProcessingStep
Training	TensorFlow/Keras	TrainingStep
Model evaluation	TFMA or custom	ProcessingStep
Deployment	SageMaker Deploy	Model/Endpoint Steps